Data-Driven Inventory Planning for a Retail Pharmacy

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled **"Data-Driven Inventory Planning for a Retail Pharmacy"**. I extend my appreciation to **Rahul Medical Store, Sambalpur**, for providing the necessary information and access that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

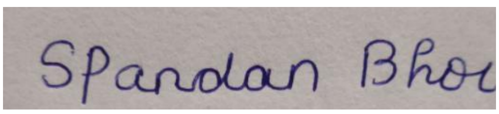
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate:



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1. Executive Summary

Rahul Medical Store, a local chemist shop in Sambalpur, Odisha, has been operating since 2008 with a focus on accessible and ethical healthcare delivery. Despite growing its inventory to over ₹90 lakhs, the store faces key challenges in managing inventory, frequent medicine expiry, and unstructured restocking. Additionally, customer reliance on non-prescription purchases and self-medication has raised accountability concerns for the pharmacy.

This project aims to address these issues by analyzing primary data including sales logs, inventory records, and expiry reports. The data will be collected through field visits and interviews with the owner and staff. Tools such as Microsoft Excel, ABC analysis, and basic forecasting methods will be employed to identify slow-moving inventory, suggest reorder levels, and reduce losses due to expired stock.

The expected outcome is a practical, data-supported inventory management model, along with recommendations to guide customer communication in cases of non-prescription medicine sales. The proposed system will help improve demand forecasting, reduce wastage, and support Rahul Medical Store in delivering more efficient and accountable service.

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2. Organization Background

Rahul Medical Store is a locally-owned, standalone retail pharmacy situated on Budharaja Road near Varun Plaza in Sambalpur, Odisha. Established in 2008 by Mr. Sanjay Kumar Pradhan, the store was founded with a mission to serve the healthcare needs of the local population. Inspired by his father's work in pharmaceutical licensing, Mr. Pradhan entered the field with a strong sense of ethics and community responsibility, prioritizing access to genuine medicines over commercial gain.

Initially launched with just ₹1 lakh worth of inventory, the store has grown significantly in the last 15 years. Today, it handles medical stock valued between ₹90 lakhs and ₹1 crore, serving hundreds of walk-in customers monthly. The store provides a broad range of products including general medicines, over-the-counter (OTC) drugs, surgical supplies, diagnostic equipment, and wellness supplements. Customers include both regular households and patients from nearby clinics, with the store earning a reputation for reliability and affordability.

Rahul Medical Store functions using conventional processes, relying heavily on Excel-based tracking and hands-on supervision by trained staff. There is no digital inventory management system in place, and orders are managed manually through supplier relationships built over the years. The owner believes in maintaining strong customer relationships through personalized service and transparency. This people-first approach has helped the store retain loyal customers even in the face of growing competition from newer stores and online platforms.

During the COVID-19 pandemic, Mr. Pradhan went out of his way to personally deliver essential medicines to customers who were unable to visit the store, reinforcing the store's commitment to service. Looking ahead, Rahul Medical Store envisions becoming a trusted and dependable healthcare partner in Sambalpur by ensuring timely access to authentic medicines and prioritizing ethical service delivery. Plans are underway to introduce elderly-friendly home delivery services and expand the store's footprint by opening additional branches across the city.

3.Problem Statement/Objectives

- **To evaluate stock movement and expiry trends to reduce wastage.**
  Unstructured stock tracking has led to overstocking and expiry of unsold medicines. This objective focuses on analyzing sales patterns and shelf-life data to identify slow-moving items and improve rotation.
- **To analyse the impact of non-prescription medicine sales on store accountability.**
  Customers often request medicines without valid prescriptions, which can lead to misuse and liability issues. This objective seeks to understand how such sales affect store reputation and explore responsible handling practices.
- **To propose a data-driven model to optimize stock management.**
  Currently, stock decisions are made manually without forecasting. This objective aims to use primary data and Excel-based tools to develop a simple reorder and tracking model that minimizes overstock and improves availability.

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4. Background of the Problem

4.1 Inventory Expiry and Overstocking

Rahul Medical Store currently relies on manual tracking through Excel and physical stock supervision by staff. While this system allows basic oversight, it lacks predictive capability. As a result, medicines are often overstocked and not sold in time, leading to expiry and wastage. The absence of stock rotation mechanisms or demand-based ordering increases the financial burden on the store due to expired inventory. Addressing this requires better data visibility and an efficient stock movement analysis.

4.2 Non-Prescription Sales and Accountability Issues

Another major challenge is the frequent sale of medicines without valid prescriptions. Many walk-in customers request medication based on internet searches or past usage, which leads to self-medication and potential misuse. Although the store follows ethical practices, it still bears the risk of being blamed for negative outcomes. Without a clear protocol to screen such customers or inform them of the risks, the store's credibility may be compromised. A solution must include responsible handling strategies.

4.3 Manual Tracking and Lack of Forecasting

The inventory system lacks automation or analytics support, making it difficult to track product demand or anticipate restocking needs. The owner and staff depend on past experience or gut feeling to decide reorder quantities. This manual approach does not scale well with growing stock volume and increases the chances of inefficiencies. A structured, data-backed system can support timely restocking and improve overall operational performance.

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5. Problem Solving Approach

a. Methods Used

This study adopts a practical, data-driven approach to solving key operational challenges faced by Rahul Medical Store. Since the store operates manually, the project will use an exploratory case study methodology, drawing insights from real-world interactions, business records, and on-ground staff behavior.

The first step involves conducting field visits and interviews with the owner and staff to understand how the inventory is currently managed, how purchase decisions are made, and how expired items are handled. These qualitative interactions will help identify bottlenecks, inconsistencies, and areas of judgment-based decision-making that may be prone to inefficiencies.

To supplement this, manual observation of restocking and customer interaction practices will be done — including how items are arranged, tracked, and sold. Patterns such as over-ordering, unsold items nearing expiry, and the absence of a structured reorder system will be documented.

Once data is collected, the following analyses will be conducted:

- **Sales Trend Analysis:** To identify frequently and rarely sold items and understand seasonal demand patterns.
- **Expiry Pattern Analysis:** To detect categories of products contributing most to expiry losses.
- **Stock Movement Review:** To measure the time from procurement to sale and identify slow-moving items.
- **ABC Classification:** To prioritize medicines based on their value contribution.
- **Forecasting Models:** To suggest reorder quantities using Excel-based time series tools.

These methods will guide recommendations to reduce wastage, improve inventory tracking, and create safe practices for handling non-prescription sales.

b. Data Collection Plan

The project will rely entirely on primary data from Rahul Medical Store. The researcher will gather both quantitative and qualitative data from:

Inventory Logs – Excel sheets and physical registers with purchase details, expiry dates, and available stock.

Sales Records – Weekly/monthly data on frequently and rarely sold medicines.

Owner/Staff Feedback – Insights on how stock is reordered, what challenges they face with non-prescription sales, and any loss due to expiry.

The specific variables that will be collected include:

- Medicine name, category, quantity, expiry date
- Purchase and selling price
- Weekly or monthly sales figures
- Reorder frequency and supplier details
- List of expired products and estimated loss
- Owner's observations about customer behavior related to non-prescription purchases

c. Analysis Tools

Microsoft Excel: The main analysis tool for building pivot tables, plotting sales and expiry trends, and calculating reorder levels.

Python: If data size and structure permit, Python will be used for descriptive statistics, plotting trends, and applying basic forecasting models using libraries like pandas and matplotlib.

Pareto Analysis: To determine the top 20% of items causing 80% of expiry losses.

ABC Classification: To group inventory items into high-value (A), moderate (B), and low-value (C) categories for focused attention.

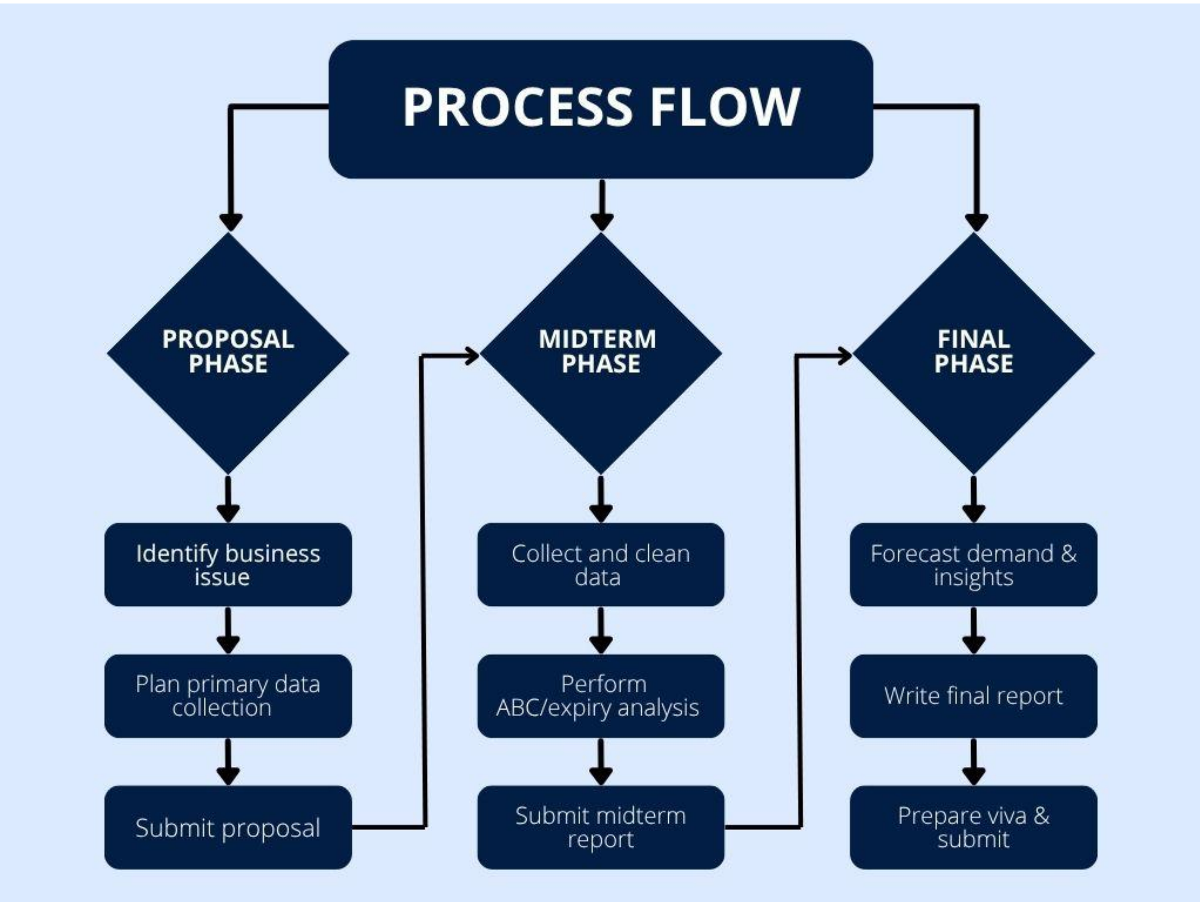
Forecasting Tools: Using Excel's built-in functions like TREND, FORECAST.LINEAR, or moving averages to estimate future demand.

Recommendation Framework: Based on the analysis, a set of practical SOPs and guidelines will be proposed to optimize restocking and handle non-prescription sales responsibly.

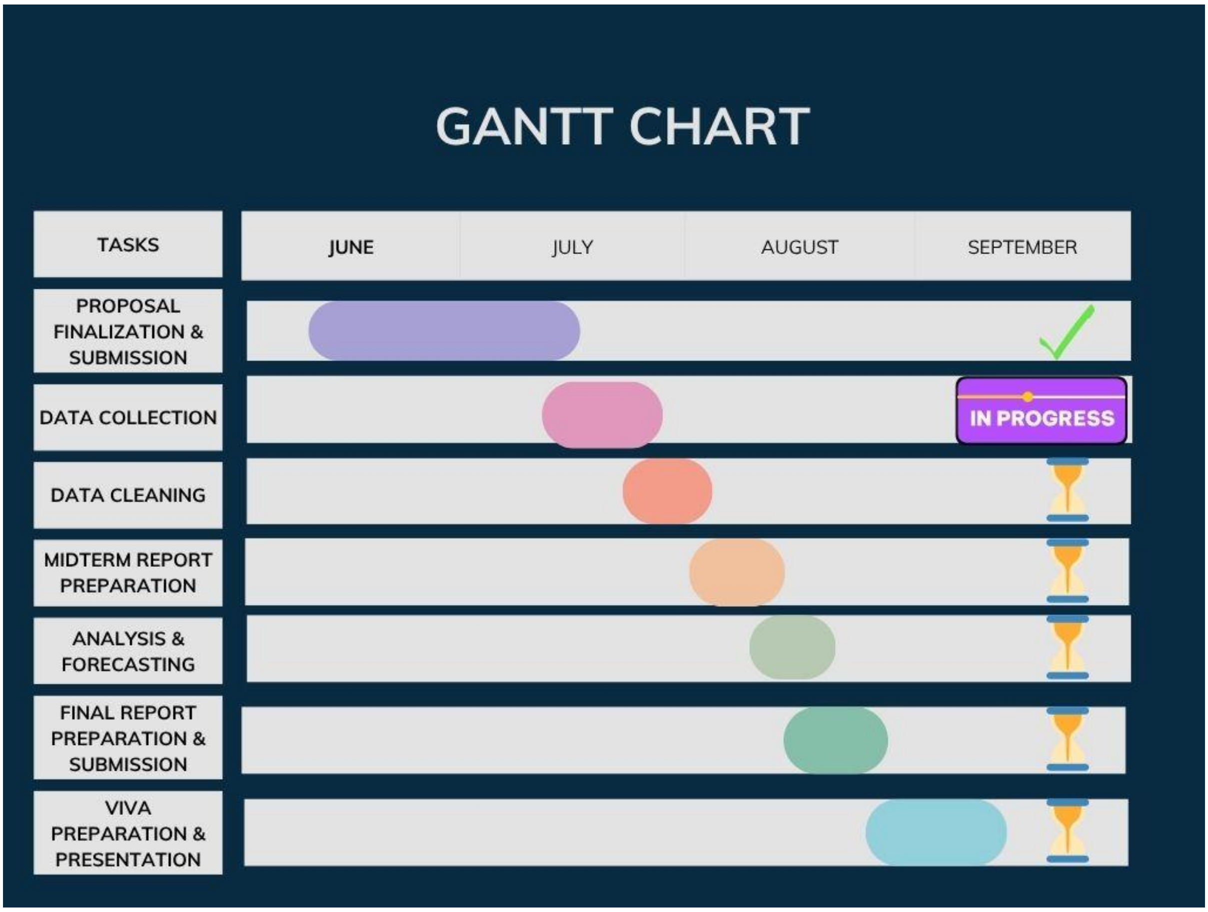
6. Expected Timeline

a. Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) below outlines the key project phases and sub-tasks. It is structured into three main phases: Proposal, Midterm, and Final. Each phase includes relevant activities that contribute toward successful completion of the BDM Capstone Project



b.The Gantt chart below illustrates the expected timeline of the BDM Capstone Project:-



7. Expected Outcomes

- Reduction in expired medicine losses by identifying slow-moving items and optimizing reorder practices
- Improved inventory tracking through a simple, structured model for forecasting and stock rotation
- Actionable recommendations for handling non-prescription sales responsibly to reduce customer-related risks

These outcomes aim to improve operational efficiency, reduce financial losses, and enhance service quality at Rahul Medical Store using data-driven insights.